

ABERPORTH, United Kingdom

WMO: 035020

Lat: 52.139N

Lon: 4.571W

Elev: 440

StdP: 14.46

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.5	31.8	21.5	16.4	32.5	24.0	18.5	34.3	46.2	44.4	42.4	45.2	12.5	150	0.763

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	71.7	61.9	68.4	60.8	65.9	59.8	63.9	68.6	62.5	66.1	61.2	64.4	11.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	85.2	65.3	61.0	81.4	63.9	59.8	78.0	62.8	29.4	68.7	28.3	66.3	27.4	64.7	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.3	33.7	30.2	DB	26.4	79.3	2.5	4.4	24.6	82.4	23.1	85.0	21.7	87.5	19.8	90.6	(4)
(5)			WB	24.3	67.8	2.5	2.3	22.4	69.4	21.0	70.7	19.5	72.0	17.7	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	42.2	42.1	44.1	47.1	51.7	56.2	59.2	59.5	56.8	52.5	47.1	43.5		
	DBStd	7.65	4.88	4.61	4.42	4.40	4.43	4.03	3.60	3.31	3.34	4.00	4.30	5.22		
	HDD50	1123	244	223	189	107	29	1	0	0	0	20	105	205		
	HDD65	5417	708	641	649	538	414	268	186	176	247	387	537	666		
	CDD50	1198	1	1	5	20	81	186	285	295	204	98	18	4		
	CDD65	16	0	0	0	0	1	3	6	5	1	0	0	0		
	CDH74	58	0	0	0	0	5	13	18	21	1	0	0	0		
	CDH80	7	0	0	0	0	0	1	2	4	0	0	0	0		
(14) Wind	WSAvg	15.0	18.7	18.0	15.8	13.9	12.7	12.1	11.7	12.6	13.9	16.0	17.1	18.1		
(15) Precipitation	PrecAvg	49.3	5.7	4.0	3.0	3.0	3.0	2.9	3.1	3.6	3.4	5.6	6.1	5.8		
	PrecMax	63.0	10.0	10.2	5.9	6.3	7.1	7.1	7.7	8.3	5.6	10.8	11.1	11.8		
	PrecMin	38.7	0.2	1.1	1.0	0.7	0.4	0.8	0.7	0.2	0.6	1.9	3.1	1.0		
	PrecStd	6.0	2.3	2.1	1.4	1.4	1.4	1.7	1.6	1.9	1.2	2.2	2.2	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	53.9	59.8	66.3	71.1	75.6	76.8	76.3	71.1	64.7	57.9	55.0		
		MCWB	51.3	50.1	50.9	55.0	59.3	64.2	64.2	64.4	62.2	58.3	55.4	53.5		
	2%	DB	51.8	51.4	54.0	59.1	65.0	69.1	70.9	70.2	66.7	61.3	56.2	53.7		
		MCWB	50.5	49.3	49.1	51.3	56.9	60.2	61.8	62.6	59.9	57.9	54.5	52.5		
	5%	DB	50.3	50.0	51.7	55.3	61.2	65.1	67.6	66.8	64.0	59.6	54.7	52.2		
		MCWB	48.8	48.1	48.7	49.9	54.8	58.5	61.1	61.1	59.3	56.9	53.0	50.8		
	10%	DB	49.0	48.6	50.1	53.0	58.1	62.2	65.1	64.5	62.0	58.2	53.3	50.8		
		MCWB	47.5	46.8	47.8	48.9	53.5	57.2	60.0	60.3	58.3	55.7	51.5	49.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	51.8	52.2	55.5	61.2	65.2	65.5	66.4	64.1	60.7	56.6	54.0		
		MCDB	52.9	52.6	55.3	64.2	68.9	73.3	72.7	73.0	68.5	62.7	57.2	54.4		
	2%	WB	50.7	50.0	50.7	52.7	58.1	61.3	63.6	64.0	61.8	59.0	55.0	52.8		
		MCDB	51.6	50.9	52.8	57.0	63.3	66.6	68.5	67.7	64.7	60.7	55.8	53.4		
	5%	WB	49.2	48.6	49.5	51.1	56.0	59.5	62.1	62.6	60.5	57.6	53.5	51.2		
		MCDB	50.2	49.6	51.2	54.4	59.7	63.6	66.0	65.2	62.7	59.2	54.5	52.1		
	10%	WB	47.8	47.1	48.1	49.7	54.3	58.1	60.8	61.2	59.2	56.2	51.9	49.6		
		MCDB	48.9	48.5	49.9	52.6	57.5	61.5	64.1	63.9	61.5	57.9	53.1	50.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.2	8.3	9.2	9.7	9.5	9.2	9.0	8.9	7.9	6.8	6.9		
		MCDBR	7.8	7.9	10.1	12.5	14.2	14.2	13.1	12.6	12.0	8.8	7.2	7.5		
	5% WB	MCWBR	8.3	7.1	7.3	7.9	8.6	8.6	7.4	7.3	7.6	6.9	7.3	8.2		
		MCWBR	7.8	7.5	9.2	11.1	12.5	12.4	11.5	10.9	9.8	8.1	7.1	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.337	0.363	0.378	0.385	0.375	0.379	0.363	0.357	0.346	0.324	0.314		
		taud	2.335	2.388	2.336	2.320	2.343	2.395	2.388	2.449	2.460	2.448	2.391	2.310		
	Ebn at Noon		211	246	262	272	275	279	275	274	262	240	211	191		
		Edh at Noon	20	25	32	36	37	36	35	32	28	23	19	17		
(44) All-Sky Solar Radiation	RadAvg		228	466	858	1362	1690	1836	1663	1393	1019	565	281	170		
	RadStd		23	42	68	95	110	155	145	92	70	47	24	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page